Assignment 3

STAT-540

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1. $dX_t = \frac{1}{2}X_t dt + X_t dW_t,\ X_0 = 1.$
2. $dX_t = \mu X_t dt + \sigma dW_t,\ X_0 = 1.$
3. $dX_t = \mu(\theta - X_t)dt + \sigma dW_t, X_0 = 1.$
4. $dX_t = -\frac{1}{2}X_t dt + \sqrt{1 - X_t^2}dW_t,\ X_0 = 1.$
5. $dX_t = \frac{b - X_t}{1-t}dt + dW_t,\ 0 \leq t < 1,\ Y_0 = a.$
6. $dX_t = -\frac{1}{1+t}X_t dt + \frac{1}{1+t}dW_t, X_0 = 0.$

Solution:

1. Problem: $dX_t = \frac{1}{2}X_t d_t + X_t dW_t\ ,\ X_0 = 1$

   Solution Method: Use Itô's formula with f(t,x) = ln(x). Let Yt = ln(Xt). By Itô's formula:

   $$dY_t = \frac{1}{X_t}dX_t - \frac{1}{2{X_t}^2}(dX_t)^2$$

   Substituting dXt:

$$dY_t = \frac{1}{X_t}\left(\frac{1}{2}X_t d_t + X_t dW_t\right) - \frac{1}{2{X_t}^2}\left(\frac{1}{2}X_t d_t + X_t dW_t\right)^2$$
$$= \frac{1}{X_t}\left(\frac{1}{2}X_t d_t + X_t dW_t\right) - \frac{1}{2{X_t}^2}\left(\frac{1}{4}\,{X_t}^2\,(d_t)^2 + \,{X_t}^2 d_t W_t + {X_t}^2 (dW_t)^2\right)$$
$$= \frac{1}{X_t}\left(\frac{1}{2}X_t d_t + X_t dW_t\right) - \frac{1}{2{X_t}^2}\left(\frac{1}{4}\,{X_t}^2\,(0) + \,{X_t}^2(0) + {X_t}^2 d_t\right)$$
$$= \frac{1}{X_t}\left(\frac{1}{2}X_t d_t + X_t dW_t\right) - \frac{1}{2{X_t}^2}\left({X_t}^2 d_t\right) = \frac{1}{2}d_t + dW_t - \frac{1}{2}d_t = dW_t$$

$$\int dY_t = \int dW_t$$

$$Y_t - Y_0 = W_t$$

Since $Y_0 = \ln(X_0) = \ln(1) = 0$:

$$\ln(Y_t) = W_t$$

$$X_t = e^{W_t}$$

2. Problem: $dX_t = \mu X_t d_t + \sigma dW_t \;,\; X_0 = 1$

Solution Method: This is a Linear SDE with additive noise. To solve it, we can use the integrating factor: e^(-μt):

Let Yt = e^(-μt)Xt. By Itô's formula:

$$dY_t = -\mu e^{-\mu t} X_t d_t + e^{-\mu t} dX_t = -\mu e^{-\mu t} X_t d_t + e^{-\mu t}(\mu X_t d_t + \sigma dW_t)$$
$$= -\mu e^{-\mu t} X_t d_t + \mu e^{-\mu t} X_t d_t + e^{-\mu t}\sigma dW_t = e^{-\mu t}\sigma dW_t$$

$$\int dY_t = \int e^{-\mu t}\sigma dW_t$$
$$Y_t - Y_0 = \sigma \int_0^t e^{-\mu s} dW_s$$

Then we sustitute Y_0, since $Y_0 = e^0 \cdot X_0 = 1$:

$$e^{-\mu t} X_t = 1 + \sigma \int_0^t e^{-\mu s} dW_s$$

$$X_t = e^{\mu t} + \sigma e^{\mu t} \int_0^t e^{-\mu s} dW_s$$

3. Problem: $dX_t = \mu(\theta - X_t)\, d_t + \sigma dW_t \ , \ X_0 = 1$

   Solution Method: This is a the Ornstein-Uhlenbeck process (mean-reverting). To solve it, we can use the integrating factor: e^(μt):

   Let Yt = e^(μt)Xt. By Itô's formula:

$$dY_t = \mu e^{\mu t} X_t d_t + e^{\mu t} dX_t = \mu e^{\mu t} X_t d_t + e^{\mu t}(\mu(\theta - X_t)d_t + \sigma dW_t)$$
$$= \mu e^{\mu t} X_t d_t + e^{\mu t}\mu\theta d_t - \mu e^{\mu t} X_t d_t + \sigma e^{\mu t} dW_t = \mu e^{\mu t}\theta d_t + \sigma e^{\mu t} dW_t$$

$$\int dY_t = \int \mu e^{\mu t}\theta d_t + \sigma e^{\mu t} dW_t$$
$$Y_t - Y_0 = \mu\theta \int_0^t e^{\mu s}\, d_s + \sigma \int_0^t e^{\mu s}\, dW_s$$

   Then we sustitute Y_0, since $Y_0 = e^0 \cdot X_0 = 1$:

$$e^{\mu t} X_t - 1 = \mu\theta \, \frac{e^{\mu t} - 1}{\mu} + \sigma \int_0^t e^{\mu s}\, dW_s$$

$$e^{\mu t} X_t = 1 + \theta(e^{\mu t} - 1) + \sigma \int_0^t e^{\mu s}\, dW_s$$
$$X_t = e^{-\mu t} + \theta(1 - e^{-\mu t}) + \sigma e^{\mu t} \int_0^t e^{\mu s}\, dW_s$$

4. Problem: $dX_t = -\frac{1}{2} X_t d_t + \sqrt{1 - {X_t}^2}\, dW_t \ , \ X_0 = 1$

   Solution Method: This equation keeps Xt bounded in [-1, 1]. To solve the problem, we can use the following transformation Xt = sin(Yt). Let Yt = arcsin(Xt), so Xt = sin(Yt).

   By Itô's formula:

$$dX_t = \cos(Y_t)\, dY_t - \frac{1}{2}\sin(Y_t)(dY_t)^2$$

   We can sustitute:

$$\cos(Y_t) = \sqrt{1 - \sin {Y_t}^2} = \sqrt{1 - {X_t}^2}$$
$$(dY_t)^2 = (diffusion\ component)^2 d_t$$

From the original SDE:

$$dX_t = -\frac{1}{2} X_t d_t + \sqrt{1 - {X_t}^2} dW_t = \sqrt{1 - {X_t}^2} dY_t - \frac{1}{2} \sin(Y_t)(dY_t)^2$$

Matching coefficients, if we try dYt = dWt:

$$\sqrt{1 - {X_t}^2} dW_t = \sqrt{1 - {X_t}^2} dY_t$$

$$-\frac{1}{2} \sin(Y_t)\, d_t = -\frac{1}{2} X_t d_t$$

Therefore dYt must equal dWt:

Integrating:

$$\int dY_t = \int dW_t$$

$$Y_t - Y_0 = W_t$$

Since $X_0 = 1$, we have $Y_0 = \arcsin(1) = \pi/2$

$$X_t = \sin\left(\frac{\pi}{2} + W_t\right) = \cos(W_t)$$

5. Problem: $dX_t = \frac{b - X_t}{1 - t} d_t + dW_t \ ,\ 0 \le t < 1, X_0 = \alpha$

Solution Method: To solve the problem, we can use the following transformation use integrating factor 1/(1-t). By Ito's formula:

$$dY_t = \frac{dX_t}{1 - t} + \frac{X_t}{(1 - t)^2}\, d_t$$

$$\frac{dX_t}{1-t} = dY_t - \frac{X_t}{(1-t)^2}d_t$$

$$\frac{b-X_t}{(1-t)^2}d_t + \frac{1}{1-t}dW_t = dY_t - \frac{X_t}{(1-t)^2}d_t$$

$$dY_t = \frac{b}{(1-t)^2}d_t + \frac{1}{1-t}dW_t$$

Integrating:

$$\int dY_t = \int \frac{b}{(1-t)^2}d_t + \frac{1}{1-t}dW_t$$

$$Y_t - Y_0 = b\int_0^t \frac{ds}{(1-s)^2} + \int_0^t \frac{1}{1-s}dW_s$$

$$\frac{X_t}{1-t} - \alpha = b\left[\frac{1}{1-s}\right]_0^t + \int_0^t \frac{1}{1-s}dW_s$$

$$\frac{X_t}{1-t} = \alpha + b\left(\frac{1}{1-t} - 1\right) + \int_0^t \frac{1}{1-s}dW_s$$

$$\frac{X_t}{1-t} = \alpha + b\left(\frac{t}{1-t}\right) + \int_0^t \frac{1}{1-s}dW_s$$

$$X_t = \alpha(1-t) + bt + (1-t)\int_0^t \frac{1}{1-s}dW_s$$

6. Problem: $dX_t = -\frac{1}{1+t}X_t d_t + \frac{1}{1+t}dW_t \ , \ X_0 = 0$

Solution Method: To solve the problem, we can use the following transformation use integrating factor (1+t). By Ito's formula:

$$dY_t = X_t d_t + (1+t)dX_t = X_t d_t + (1+t)\left(-\frac{1}{1+t}X_t d_t + \frac{1}{1+t}dW_t\right)$$
$$= X_t d_t + (-X_t d_t + dW_t) = dW_t$$

Integrating:

$$\int dY_t = \int dW_t$$

$$Y_t - Y_0 = W_t$$

Since $Y_0 = (1+0)X_0 = 0$:

$$X_t(1+t) = W_t$$

$$X_t = \frac{W_t}{(1+t)}$$